LAYERED PERMUTATION COPULAS ARE CLOSER TO THE UNIFORM LAW THAN THE DIAGONAL

YITZCHAK SHMALO

Abstract. Schrott conjectured that among all copulas on the unit square, the diagonal and the antidiagonal couplings are the farthest from the uniform law in the quadratic Wasserstein distance, at squared distance 1/10. By density of permutation copulas the conjecture is implied by a statement about n equally weighted grid points: for every permutation σ of $\{0,\dots,n-1\}$, the points $\big((i+\frac{1}{2})/n,(\sigma(i)+\frac{1}{2})/n\big)$ are no farther from the uniform law than the diagonal points. We prove this, with strict inequality, for every layered permutation, that is, every permutation obtained by cutting $0,\dots,n-1$ into runs of consecutive values and reversing some of the runs, other than the identity and the full reversal. The proof re-cuts the optimal partition of the identity on the union of the moved strips, writes the gain as a weighted sum of two-cell comparisons, and settles each comparison by a dispersive-order inequality between the laws of x_1+x_2 and x_1-x_2 on a diagonal band. The same re-cut, done in two steps inside a window of three or four strips, gives the strict inequality for every permutation that splits into blocks of consecutive values of length at most four (each block reversed or permuted arbitrarily), once $n\ge 9$; for blocks of length four the positivity of six explicit functionals is proved with interval arithmetic.

1. The problem

Let λ be Lebesgue measure on $Q=[0,1]^2$. A copula is a probability measure on Q with uniform marginals. Schrott [1] proved that among all couplings of n standard Gaussians, the ones farthest from the standard Gaussian on $\mathbb{R}^n$ in W_2 are the signed diagonals, and stated the uniform analogue as a conjecture.

Conjecture 1.1 ([1, Conjecture 1.1])**.** *For every copula π on Q, $W_2^2(\pi,\lambda)\le 1/10$, with equality for the diagonal coupling.*

For a probability measure μ on Q write $V(\mu)=\max\mathbb{E}[X\cdot Z]$ over couplings of $X\sim\lambda$ and $Z\sim\mu$. Since $W_2^2(\mu,\lambda)=\mathbb{E}|X|^2+\mathbb{E}|Z|^2-2V(\mu)$ and $\mathbb{E}|Z|^2=2/3$ for every copula, Conjecture 1.1 says that $V(\pi)\ge V(\text{diagonal})$ for every copula π.

Fix $n\ge 2$, put $a_i=(i+\frac{1}{2})/n$ for $0\le i\le n-1$, and for a permutation σ of $\{0,\dots,n-1\}$ let $\mu_\sigma=\frac{1}{n}\sum_i\delta_{p_i}$ with $p_i=(a_i,a_{\sigma(i)})$, and $A_n(\sigma)=V(\mu_\sigma)$. Both marginals of μ_σ are the uniform law on the grid $\{a_i\}$, so $\mathbb{E}|Z|^2$ does not depend on σ, and $A_n(\sigma)\ge A_n(\mathrm{id})$ says that μ_σ is no farther from λ than the diagonal points. Consider

$$(\mathrm{A}_n)\qquad A_n(\sigma)\ge A_n(\mathrm{id})\qquad\text{for every permutation }\sigma\text{ of }\{0,\dots,n-1\}.$$

Every copula π is a weak limit of measures μ_{σ_n}. Indeed, let $Z^1,\dots,Z^n$ be independent with law π, almost surely with distinct coordinates, and let σ_n send the rank of Z_1^j to the rank of Z_2^j (ranks counted from 0). Replacing each coordinate by $(\mathrm{rank}+\frac{1}{2})/n$ moves it by at most $\sup_t|F_n(t)-t|+\frac{1}{2n}$, where F_n is the empirical distribution function of that coordinate, which tends to 0 almost surely by the Glivenko–Cantelli theorem; and the empirical measure of the Z^j tends weakly to π. On Q weak convergence is W_2 convergence, and $|V(\mu)-V(\mu')|\le(\mathbb{E}|X|^2)^{1/2}W_2(\mu,\mu')$. Hence, if

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(A_n) holds for all n, then $V(\pi) = \lim_n A_n(\sigma_n) \geq \lim_n A_n(\mathrm{id}) = V(\text{diagonal}) = 37/60$ for every copula π, the last limit by the same argument applied to the diagonal. So (A_n) for all n implies Conjecture 1.1.

The map A_n is invariant under the eight symmetries of the square acting on permutation matrices, so $A_n(\mathrm{rev}) = A_n(\mathrm{id})$ for the full reversal $\mathrm{rev}(i) = n-1-i$. Numerically (A_n) holds for every permutation when $n \leq 8$, with equality only at id and rev (Section 5).

Definition 1.2. A permutation σ is *layered* if there are integers $0 = k_0 < k_1 < \cdots < k_m = n$ such that, for each ℓ, σ maps $\{k_\ell, \ldots, k_{\ell+1} - 1\}$ onto itself either identically or by reversal.

Theorem 1.3. *Let $n \geq 2$ and let σ be a layered permutation of $\{0, \ldots, n-1\}$ other than* id *and* rev. *Then $A_n(\sigma) > A_n(\mathrm{id})$. The same holds for the images of σ under the symmetries of the square.*

For $n = 2$ every permutation is id or rev, so the theorem is about $n \geq 3$. Section 4 extends Theorem 1.3 to permutations built from blocks of length at most four (Theorem 4.5).

2. The identity and the re-cut

Write $s = x_1 + x_2$ and $v = x_1 - x_2$ for $x \in Q$. Under λ, $S = X_1 + X_2$ has density $L(s) = \min(s, 2-s)$ on $[0,2]$, and given $S = s$ the variable $V = X_1 - X_2$ is uniform on $[-L(s), L(s)]$. Let $0 = q_0 < q_1 < \cdots < q_n = 2$ be the quantiles of S at levels j/n and $T_j = \{x \in Q : q_j < s < q_{j+1}\}$.

Lemma 2.1. $A_n(\mathrm{id}) = \sum_j a_j \int_{T_j} s\,dx$.

Proof. A coupling of λ with μ_σ gives each point p_i a sub-probability $\gamma_i \leq \lambda$ of mass $1/n$, and $\mathbb{E}[X \cdot Z] = \sum_i \int x \cdot p_i \, d\gamma_i$. For $\sigma = \mathrm{id}$, $x \cdot p_j = a_j s$, and with $a_0 < \cdots < a_{n-1}$ the sum $\sum_j a_j \int s\, d\gamma_j$ under $\sum_j \gamma_j = \lambda$, $\gamma_j(Q) = 1/n$, is largest when γ_j is λ on T_j (the rearrangement inequality applied to the law of S). □

Any partition of Q into regions $R_0, \ldots, R_{n-1}$ of area $1/n$ gives the coupling $\gamma_i = \lambda|_{R_i}$, so $A_n(\sigma) \geq \sum_i \int_{R_i} x \cdot p_i \, dx$.

Consider a block of $r \geq 2$ consecutive values $k, \ldots, k+r-1$ with $k + r \leq n$, and let σ reverse it and fix every other value. Its moved points are $p_{k+i} = (a_{k+i}, a_{k+r-1-i})$, $0 \leq i \leq r-1$; they share $p_1 + p_2 = 2\bar{a}$ with $\bar{a} = a_k + (r-1)/(2n)$, and $p_1 - p_2 = w_i := (2i+1-r)/n$, increasing in i. Keep T_j for every fixed point, and cut $U = T_k \cup \cdots \cup T_{k+r-1} = \{q_k < s < q_{k+r}\}$ into the regions $R_0, \ldots, R_{r-1}$ of area $1/n$ cut out by consecutive quantiles of v on U, from the smallest values of v up; give R_i to p_{k+i}. Since $x \cdot p = \frac{1}{2}\big(s(p_1+p_2) + v(p_1 - p_2)\big)$, Lemma 2.1 gives $A_n(\sigma) - A_n(\mathrm{id}) \geq D$ with

$$D = \sum_{i=0}^{r-1} \int_{R_i} \Big(\bar{a}s + \tfrac{1}{2} w_i v\Big) dx - \sum_{j=k}^{k+r-1} a_j \int_{T_j} s\, dx = \frac{1}{n^2}\Big(\sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\, \nu_i - \sum_{i=0}^{r-1} (i - \tfrac{r-1}{2})\, m_i\Big),$$

where $m_i = n \int_{T_{k+i}} s\, dx$ is the mean of S on the strip T_{k+i} and $\nu_i = n \int_{R_i} v\, dx$ the mean of V on R_i.

Lemma 2.2. *For real numbers $c_0, \ldots, c_{r-1}$, $\sum_i (i - \frac{r-1}{2}) c_i = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}\, \Delta_t(c)$ with $\Delta_t(c) = \frac{1}{r-t} \sum_{i \geq t} c_i - \frac{1}{t} \sum_{i < t} c_i$.*

Proof. With $\bar{c}$ the mean, $\sum_i i(c_i - \bar{c}) = \sum_{t \geq 1} \sum_{i \geq t} (c_i - \bar{c})$, and $\sum_{i \geq t} (c_i - \bar{c}) = (r-t)\big(\frac{1}{r-t} \sum_{i \geq t} c_i - \bar{c}\big) = \frac{t(r-t)}{r} \Delta_t(c)$. □

So $n^2 D = \sum_{t=1}^{r-1} \frac{t(r-t)}{r}(\Delta_t(\nu) - \Delta_t(m))$. Both ν and m are the means of a variable over its own quantile cells of mass $1/n$ on U: $\Delta_t(\nu)$ is the gap between the mean of V above and below its quantile at mass t/n on U, and $\Delta_t(m)$ the same gap for S.

3. A dispersive comparison on diagonal bands

Fix $0 \le \alpha < \beta \le 2$ with $(\alpha, \beta) \ne (0, 2)$ and let $U = \{\alpha < s < \beta\}$, of area ρ. Let Φ_S and Φ_V be the distribution functions of S and V under $\lambda|_U$ (total mass ρ), Q_S, Q_V their inverses on $(0, \rho)$, and $\phi_S = L \circ Q_S$, $\phi_V = g \circ Q_V$ the densities read at mass level μ, where $g(y) = \frac{1}{2}|(\alpha, \beta) \cap (|y|, 2 - |y|)|$ is the density of V under $\lambda|_U$. Write $\mu' = \rho - \mu$.

Lemma 3.1. *For $0 < \mu < \rho$, $\phi_V(\mu) \le \sqrt{2\min(\mu, \mu')} \le \phi_S(\mu)$. If $\alpha > 0$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $0 < \mu < c$, and if $\beta < 2$ there is $c > 0$ with $\phi_V(\mu) < \phi_S(\mu)$ for $\rho - c < \mu < \rho$.*

Proof. Let $s = Q_S(\mu)$. If $s \le 1$, then $\mu = \int_\alpha^s L \le \int_0^s t\, dt = s^2/2$, so $\phi_S(\mu) = s \ge \sqrt{2\mu}$, strictly if $\alpha > 0$. If $s \ge 1$, then $\mu' = \int_s^\beta L \le (2 - s)^2/2$ and $\phi_S(\mu) = 2 - s \ge \sqrt{2\mu'}$, strictly if $\beta < 2$. In both cases $\phi_S(\mu) \ge \sqrt{2\min(\mu, \mu')}$.

For the left inequality, g is even. For $y \in [-1, 0]$ the interval $(-y, 2 + y)$ grows with y and each of its ends moves at unit speed, so g is nondecreasing on $[-1, 0]$, $g(-1) = 0$, and $g(y) - g(z) \le y - z$ for $-1 \le z \le y \le 0$. Hence for $y = Q_V(\mu) \le 0$,

$$\mu = \int_{-1}^{y} g(z)\, dz \ge \int_{-1}^{y} \big(g(y) - (y - z)\big)_+ dz \ge \tfrac{1}{2} g(y)^2,$$

so $\phi_V(\mu) \le \sqrt{2\mu}$, and $\mu \le \mu'$ by the symmetry of g. The case $y \ge 0$ is the mirror image.

For the strict inequality let $\alpha > 0$. If $\alpha < 1$, put $c = \min(\Phi_S(1), \rho) > 0$: for $\mu < c$ we have $Q_S(\mu) < 1$ and $\phi_S(\mu) > \sqrt{2\mu} \ge \phi_V(\mu)$ by the first paragraph. If $\alpha \ge 1$, put $c = \rho/2$: then $Q_S(\mu) \ge 1$, and for $\mu < c$ we have $\mu < \mu'$, so $\phi_S(\mu) \ge \sqrt{2\mu'} > \sqrt{2\mu} \ge \phi_V(\mu)$. The case $\beta < 2$ follows by applying the map $(x_1, x_2) \mapsto (1 - x_2, 1 - x_1)$, which preserves λ, fixes v, sends s to $2 - s$ and the band (α, β) to $(2 - \beta, 2 - \alpha)$, and reverses the mass levels of S. □

Proposition 3.2. *Let $n \ge 3$, $2 \le r \le n - 1$, $0 \le k \le n - r$. Then $D > 0$ for the reversal of the block $k, \dots, k + r - 1$.*

Proof. Take $\alpha = q_k$, $\beta = q_{k+r}$; since $r < n$, $(\alpha, \beta) \ne (0, 2)$. Since $Q_S' = 1/\phi_S$ and $Q_V' = 1/\phi_V$, Lemma 3.1 shows that $h = Q_V - Q_S$ is nondecreasing on $(0, \rho)$. For $\tau = t/n$,

$$\Delta_t(\nu) - \Delta_t(m) = \frac{1}{\tau(\rho - \tau)} \int_0^\tau \int_\tau^\rho \big(h(\mu) - h(\mu_0)\big)\, d\mu\, d\mu_0 \ge 0.$$

If $\alpha > 0$, Lemma 3.1 gives $c > 0$ with h strictly increasing on $(0, c)$, so $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \min(\tau, c)/2$ and $\mu > \tau$; the integral is then positive for every t. If $\alpha = 0$ then $\beta < 2$, h is strictly increasing on some $(\rho - c, \rho)$, and $h(\mu) > h(\mu_0)$ whenever $\mu_0 < \tau$ and $\mu > \rho - \min(\rho - \tau, c)/2$. By Lemma 2.2, $n^2 D > 0$. □

Proof of Theorem 1.3. Let the reversed runs of σ be $B_1, \dots, B_m$, of sizes $r_\ell \ge 2$. If some $r_\ell = n$ then $\sigma = \mathrm{rev}$; otherwise each $r_\ell \le n - 1$, and $m \ge 1$ since $\sigma \ne \mathrm{id}$. The unions U_ℓ of the strips of the runs are disjoint, so performing the re-cut of Section 2 inside every U_ℓ and keeping the other strips is a partition of Q into regions of area $1/n$ matched to the points of μ_σ. Its value is $A_n(\mathrm{id}) + \sum_\ell D_\ell$, and each $D_\ell > 0$ by Proposition 3.2. The symmetries of the square preserve λ and the grid, so A_n is invariant under them. □

Remark 3.3. For $r = 2$ at the corner ($k = 0$ or $k = n - 2$, $n \ge 4$) the bound is exactly $n^2 D = \frac{2\sqrt{2} - 2}{3}\, n^{-1/2} \approx 0.276\, n^{-1/2}$, while the measured gap $n^2(A_n(\sigma) - A_n(\mathrm{id}))$ is about $0.344\, n^{-1/2}$ ($n = 8, 12, 16$). The corner transposition is the cheapest move, so the margin in (A_n) near the identity is of order $n^{-5/2}$.

4. Windows of three and four values

Let $r \geq 2$, let τ be a permutation of $\{0, \dots, r-1\}$, $0 \leq k \leq n-r$, and let $\sigma(k+i) = k + \tau(i)$ for $0 \leq i \leq r-1$, σ the identity elsewhere. Put $u_i' = i + \tau(i) - (r-1)$ and $v_i' = i - \tau(i)$. The moved points satisfy $p_1 + p_2 = 2\bar{a} + u_i'/n$ and $p_1 - p_2 = v_i'/n$ for $p = p_{k+i}$, with $\bar{a}$ and $U = \{q_k < s < q_{k+r}\}$ as in Section 2. For a region C of positive area write $\mathbb{E}_C$ for the average over C with respect to λ.

Lemma 4.1. *Let $C_0, \dots, C_{r-1}$ be a partition of U into regions of area $1/n$, and*

$$F = \sum_{i=0}^{r-1} (u_i' \, \mathbb{E}_{C_i}[S] + v_i' \, \mathbb{E}_{C_i}[V]) - \sum_{j=0}^{r-1} (2j - r + 1) \, \mathbb{E}_{T_{k+j}}[S].$$

Then $A_n(\sigma) - A_n(\mathrm{id}) \geq F/(2n^2)$. If σ acts in this way on several disjoint blocks, re-cutting each block's strips separately gives $A_n(\sigma) - A_n(\mathrm{id}) \geq \sum_\ell F_\ell/(2n^2)$.

Proof. Give C_i to p_{k+i} and keep T_j for the fixed points. With $x \cdot p = \frac{1}{2}(s(p_1 + p_2) + v(p_1 - p_2))$ and $\int_C g \, dx = \frac{1}{n}\mathbb{E}_C[g]$, $\sum_i \int_{C_i} x \cdot p_{k+i} \, dx = \bar{a} \int_U s \, dx + \frac{1}{2n^2} \sum_i (u_i' \mathbb{E}_{C_i}[S] + v_i' \mathbb{E}_{C_i}[V])$, while by Lemma 2.1 the identity collects $\sum_j a_{k+j} \int_{T_{k+j}} s \, dx = \bar{a} \int_U s \, dx + \frac{1}{2n^2} \sum_j (2j - r + 1) \mathbb{E}_{T_{k+j}}[S]$. Disjoint blocks use disjoint unions of strips, so the gains add. □

For the reversal of the block, $u_i' = 0$ and $v_i' = 2i - r + 1$, and $F/(2n^2)$ is the quantity D of Section 2. The strips T_{k+j} are the cells of U between consecutive quantiles of S. We use two partitions. *v then s*: group the indices i by the value of v_i'; cut U by V at its quantiles of mass $1/n$ times the cumulative group sizes, the lowest values of V going to the group with the smallest v'; cut each group's region by S into cells of area $1/n$ and give them to the group's indices in increasing order of u_i'. *s then v* is the same with the roles of (S, u') and (V, v') exchanged. The band U is symmetric under $v \mapsto -v$, so the median of V on U is 0.

The symmetries $(x_1, x_2) \mapsto (x_2, x_1)$ and $(x_1, x_2) \mapsto (1 - x_2, 1 - x_1)$ preserve λ, map strips to strips and windows to windows, and replace τ by τ^{-1} and by $i \mapsto r - 1 - \tau^{-1}(r - 1 - i)$; their composite gives $i \mapsto r - 1 - \tau(r - 1 - i)$. The second sends the band $\{\alpha < s < \beta\}$ to $\{2 - \beta < s < 2 - \alpha\}$. The image of a partition is a partition with the same value of F. So it suffices to treat one block in each orbit, on every band.

Proposition 4.2. *Let $n \geq 4$ and $\tau = (1, 2, 0)$ or $(2, 0, 1)$. Then the v then s partition gives $F > 0$ on every band U of three consecutive strips.*

Proof. The two 3-cycles form one orbit; take $\tau = (1, 2, 0)$, with $(u_i', v_i') = (-1, -1), (1, -1), (0, 2)$. Let c be the upper tercile of V on U (so $c \geq 0$), $C_2 = U \cap \{v > c\}$, and split the rest $U' = U \setminus C_2$ at the median of S on U' into C_0 (lower) and C_1 (upper). Since $\mathbb{E}_U[V] = 0$, $\mathbb{E}_{C_0}[V] + \mathbb{E}_{C_1}[V] = 2\mathbb{E}_{U'}[V] = -\mathbb{E}_{C_2}[V]$, and

$$F = 3 \, \mathbb{E}_{C_2}[V] + (\mathbb{E}_{C_1}[S] - \mathbb{E}_{C_0}[S]) - 2(\mathbb{E}_{T_{k+2}}[S] - \mathbb{E}_{T_k}[S]) = \tfrac{3}{2}(G_V - G_S) + (\Delta_{U'}(S) - \tfrac{1}{2} G_S),$$

where G_X is the mean of X on the upper third of U (by mass) minus its mean on the lower third, and $\Delta_{U'}(S)$ the gap between the means of S above and below its median on U'. With $h = Q_V - Q_S$ nondecreasing (Lemma 3.1), $G_V - G_S = \frac{3}{\rho}(\int_{2\rho/3}^{\rho} h - \int_0^{\rho/3} h) \geq 0$. For the second bracket, the section of U' over s has length $L(s) + \min(L(s), c) \geq L(s)$, half the section of U. For any law, the gap between the means above and below the median is $2 \min_t \mathbb{E}|X - t|$; the mass of U' is $2/3$ that of U, so $\Delta_{U'}(S) \geq \frac{3}{4}\Delta_U(S)$. The upper half of U is its upper third together with the upper half of its middle third, in proportions $2 : 1$, and likewise below, so $\Delta_U(S) > \frac{2}{3} G_S$. Hence $\Delta_{U'}(S) - \frac{1}{2} G_S > 0$ and $F > 0$. □

For $r = 4$ there are thirteen indecomposable permutations (no proper initial segment $\{0, \dots, j-1\}$ is mapped onto itself); one is the reversal and the other twelve fall into six orbits,

$$\{1302, 2031\},\ \{1320, 2130, 3021, 3102\},\ \{3120\},\ \{2301\},\ \{2310, 3201\},\ \{1230, 3012\},$$

writing τ as the word $\tau(0)\tau(1)\tau(2)\tau(3)$. On a band U let $S_1, \dots, S_4$ be the means of S on the four strips, $d_{41} = S_4 - S_1$, $d_{32} = S_3 - S_2$, x_m the median of $|V|$ on U, A and B the means of $|V|$ above and below x_m, T_{out} and T_{in} the means of S on $\{|v| > x_m\}$ and $\{|v| < x_m\}$, and R_1, R_3 the means of S on the lowest and highest thirds (by mass, in s) of $\{v < x_m\}$. By the symmetry $v \mapsto -v$ the quartiles of V on U are $-x_m, 0, x_m$.

Lemma 4.3. *The v then s partition gives*

$$\begin{aligned}
F(1302) &= 4A + 2B - 3d_{41} - d_{32}, & F(1320) &= 5A + B + (T_{\text{out}} - T_{\text{in}}) - 3d_{41} - d_{32},\\
F(2301) &= 4A + 4B - 2d_{41}, & F(2310) &= 5A + 3B - \tfrac{5}{2}d_{41} - \tfrac{1}{2}d_{32},\\
F(1230) &= 4A + 2(R_3 - R_1) - 3d_{41} - d_{32}, & F(3120) &= 6A + (T_{\text{hi}} - T_{\text{lo}}) - 3d_{41} - d_{32},
\end{aligned}$$

where $T_{\text{hi}}, T_{\text{lo}}$ are the means of S above and below its median on $\{|v| < x_m\}$.

Proof. For 1302 the points are $(u', v') = (-2, -1), (1, -2), (-1, 2), (2, 1)$; the v' are distinct, so the cells are the quartile cells of V, $\{v < -x_m\}, \{-x_m < v < 0\}, \{0 < v < x_m\}, \{v > x_m\}$, carrying $v' = -2, -1, 1, 2$ and $u' = 1, -2, 2, -1$. The V-means are $-A, -B, B, A$, which gives $4A + 2B$; the S-means are $T_{\text{out}}, T_{\text{in}}, T_{\text{in}}, T_{\text{out}}$, and the weights $1 - 1$ and $-2 + 2$ cancel them. The strips contribute $-(3S_4 + S_3 - S_2 - 3S_1) = -3d_{41} - d_{32}$. The other five are the same computation. For 1320, $(u', v') = (-2, -1), (1, -2), (1, 0), (0, 3)$, the cells are again the V-quartiles. For 2301 the points are $(\pm 1, \pm 2)$: the halves $\{v < 0\}$, $\{v > 0\}$ are split at the S-median, which is that of U, and the V-means add to $2(A + B)$ on each side. For 2310, $(u', v') = (-1, -2), (1, -2), (0, 1), (0, 3)$, the half $\{v < 0\}$ is split by S and the upper half into $\{0 < v < x_m\}$ and $\{v > x_m\}$. For 1230, $(u', v') = (-2, -1), (0, -1), (2, -1), (0, 3)$, the top quarter is $\{v > x_m\}$ with V-mean A, the rest $\{v < x_m\}$ has V-mean $-A/3$ and is cut into S-thirds. For 3120, $(u', v') = (0, -3), (-1, 0), (1, 0), (0, 3)$, the outer quarters go to $v' = \pm 3$ and $\{|v| < x_m\}$ is split at its S-median. □

Proposition 4.4. *Let τ be one of the twelve indecomposable non-layered permutations of $\{0, 1, 2, 3\}$. On every band U of mass $\rho \le 4/9$, the v then s partition gives $F > 0$.*

Proof. By the orbits it suffices to treat the six representatives. Always $A \ge x_m \ge B \ge 0$, $T_{\text{hi}} > T_{\text{lo}}$, $R_3 > R_1$ and $d_{41} > d_{32} > 0$. Lemma 4.3 gives

$$\begin{aligned}
F(3120) - F(1302) &= 2(A - B) + (T_{\text{hi}} - T_{\text{lo}}), & F(2301) - F(1302) &= 2B + d_{41} + d_{32},\\
F(2310) - F(1302) &= A + B + \tfrac{1}{2}(d_{41} + d_{32}), & F(1230) &\ge F_0 := 4A - 3d_{41} - d_{32},
\end{aligned}$$

all differences positive, and $F(1302) = F_0 + 2B \ge F_0$. So it remains to show $F_0 > 0$ and $F(1320) > 0$ for $\rho \le 4/9$. Write $U = \{\alpha < s < \beta\}$ and $w = \beta - \alpha$.

Thin bands across the diagonal, $\alpha < 1 < \beta$, $w < 1/3$. On U, $L \ge 1 - w$, so $\rho \ge w(1 - w)$. The distribution function of $|V|$ on U is $H(x) = \int_\alpha^\beta \min(L(s), x)\, ds \le wx$, and $H(x_m) = \rho/2$, so $x_m \ge (1 - w)/2$ and $A \ge (1 - w)/2$. All S_j, T and R lie in (α, β), so each difference of them is at most w in absolute value. Hence $F_0 \ge 2(1 - w) - 4w > 0$ and $F(1320) \ge \frac{5}{2}(1 - w) - 5w > 0$.

Bands on one side of the diagonal, $\beta \le 1$ (the case $\alpha \ge 1$ follows by $s \mapsto 2 - s$, which leaves A, B, d_{41}, d_{32} unchanged and reverses the sign of $T_{\text{out}} - T_{\text{in}}$). Every quantity in the functionals is a length, and the band $\{\alpha < s < \beta\}$ is the band $\{\alpha/\beta < s < 1\}$ scaled by β, so each functional equals $\beta G(\alpha/\beta)$ for an explicit algebraic function G on $[0, 1)$: the quantiles of S and of $|V|$ solve quadratic equations, and the means are ratios of polynomial integrals. We enclosed G

on subintervals of $[0,1]$ in interval arithmetic with outward rounding (`mpmath.iv`), bisecting until the lower end of the enclosure is positive: F_0 in 24 boxes (smallest lower bound 0.035), $F(1320)$ in 12 boxes on $\beta \le 1$ (0.124) and 17 boxes on $\alpha \ge 1$ (0.172). Since this covers every $t = \alpha/\beta \in [0,1]$, it holds for bands of any mass.

Wider bands across the diagonal, $\alpha < 1 < \beta$, $\rho \le 4/9$, $w \ge w_0$. In the coordinates $(\alpha, 2-\beta) \in [0,1]^2$, $\rho = (2 - \alpha^2 - (2-\beta)^2)/2$, and ρF is a combination of the integrals $K(x) = \int_0^x sL(s)\,ds$ at α, β and the quartiles of S, of $N(x) = \int_0^x y\,dH(y)$ at x_m and 1, and, for 1320, of $P(x_m) = \int_\alpha^\beta s\,(L(s) - x_m)_+\,ds$, which gives T_{out}. The quartiles of S increase in α and β, K and N increase and P decreases in x, and x_m is enclosed by bisection on the sign of $H(x) - \rho/2$ over a whole box; so each box of parameters gets an enclosure of ρF. Bisecting boxes until the enclosure is positive: F_0 with $w_0 = 0.05$ in 2,774 boxes, $F(1320)$ with $w_0 = 0.25$ in 1,042 boxes, no failures. Together with the thin case this covers every band of mass at most $4/9$. □

Theorem 4.5. *Let $n \ge 9$, and let $\sigma \ne \mathrm{id}, \mathrm{rev}$ map each interval $\{k_\ell, \dots, k_{\ell+1} - 1\}$ of a cut $0 = k_0 < \cdots < k_m = n$ onto itself, acting on each interval either by reversal or by a permutation of an interval of length at most four. Then $A_n(\sigma) > A_n(\mathrm{id})$, and the same holds for the images of σ under the symmetries of the square. If every interval of length four is reversed or fixed, $n \ge 4$ suffices.*

Proof. Refine the cut so that σ is indecomposable on each interval. An indecomposable piece of length at most four is a fixed point, a reversal, one of the two 3-cycles, or one of the twelve blocks of Proposition 4.4; a reversed piece of length n would make $\sigma = \mathrm{rev}$. By Lemma 4.1 the gain is at least the sum over the moved pieces of $F_\ell/(2n^2)$, and each F_ℓ is positive: reversals by Proposition 3.2, 3-cycles by Proposition 4.2, the twelve blocks by Proposition 4.4, since four strips have mass $4/n \le 4/9$. □

For $n \le 8$ every permutation was checked numerically (Section 5). The blocks 1302 and 1320 need the restriction on the mass: their v then s value is negative on nearly full bands, $\{0 < s < 1.8\}$ for 1302 and $\{0.2 < s < 2\}$ for 1320.

5. Computations

All computations use exact moments of S and one-dimensional quadrature of the law of V on a band.

(1) Check of (A_n) over every permutation for $n \le 8$ (one per class under the symmetries of the square; 5,282 classes at $n = 8$), with A_n computed by a smoothed semi-discrete dual on a 40×40 grid of cells. No permutation falls below the identity; the smallest positive gap is the corner adjacent transposition ($n^2(A_n(\sigma) - A_n(\mathrm{id})) = 0.1215$ at $n = 8$), and the first non-layered permutations are corner 3-cycles (0.3944 at $n = 8$).
(2) The bound D for every block reversal, $3 \le n \le 48$, $2 \le r \le n-1$, every k: all positive, with minimum at the corner adjacent transposition, $n^{5/2}D = 0.2761$, matching the closed form above. Quadrature error below 10^{-8}.
(3) The identity of Lemma 2.2 applied to D, checked against the direct evaluation for $n = 5, 8, 13$ and every (k, r): difference below 1.1×10^{-8}; every term $\Delta_t(\nu) - \Delta_t(m)$ is positive.
(4) The closed forms of Lemma 4.3 against a direct evaluation of the v then s cut on polygons, on seven bands: agreement to 10^{-7}; the values agree across each orbit to 5×10^{-15}.
(5) The interval enclosures of Proposition 4.4, in `mpmath.iv`; the scripts are in the repository. As an independent check, $F(1302)$ was also proved positive directly: 17 boxes on one-sided bands (smallest lower bound 0.129) and 1,550 boxes on crossing bands of mass at most $4/9$.

6. What remains

Theorems 1.3 and 4.5 cover the permutations that split into layers and blocks of at most four values. Inside a window of five strips a directional cut still gives positive bounds for every indecomposable block that was tested ($n = 8$), but no closed form of such a cut is known for blocks of five or more values. On the whole square a cut along one direction is not enough: cutting Q into quantile cells of $s\cos\theta + v\sin\theta$ and matching the cells optimally never beats the strips for any θ when $n = 4$. A proof of (A_n) along these lines would need cuts whose direction varies across the square. For $n \le 8$ the inequality rests on the computation of Section 5.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the reduction, the re-cut construction and the proofs, wrote and ran the computations of Section 5, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Sections 4 and 5, including the interval-arithmetic proofs, with its output, is available at `https://github.com/yspennstate/wasserstein-layered-permutation-copulas`.

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com